ALEXANDRU IOAN CUZA UNIVERSITY of IAȘI

FACULTY OF COMPUTER SCIENCE

ECODAM 2026

Programme – a chronological glance

June 17

15:00 Registration – *Ferdinand* Hall
16:30 Session 5 – *Ferdinand Hall*
18:30 Welcome Dinner – *Panoramic Restaurant* (*Unirea* Hotel)

June 18 – *Ferdinand* Hall (see map on page 7)

9:00 Registration
9:30 Opening Session
10:15 Session 2
15:00 Session 3
16:15 Students' 3-minute presentations

June 19 – *Ion Simionescu* Hall (see map on page 7)

9:30 Session 4
10:45 Session 6
15:00 Session 7
17:15 Closing Session

Programme – a track-based glance

ECO-Mining track

Evolutionary Computing, Optimisation and Data Mining

June 18, *Ferdinand Hall*

10:15 Session 2
15:00 Session 3

June 19, *Ion Simionescu* Hall

9:30 Session 4

BioInformatics /Applied Machine Learning track

June 17, *Ferdinand* Hall

16:30 Session 5

June 19, *Ion Simionescu* Hall

10:45 Session 6
15:00 Session 7

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ECODAM 2026

FACULTY OF COMPUTER SCIENCE

Programme

Registration: June 17, 15:00; June 18, 9:00 – *Ferdinand* Hall

Opening Session: **21 years with ECODAM**
June 18, 9:30 – *Ferdinand* Hall

10:00 – 10:15 Coffee break

ECO-Mining track

Evolutionary Computing, Optimisation and Data Mining
June 18, *Ferdinand* Hall

Session 2

10:15 – 11:15 **Dan Simovici** (University of Massachusetts, Boston): *"Compressibility, Mineability and Large Language Models"*

11:15 – 12:15 **Darrell Whitley** (Colorado State University): "*Evolutionary Optimization, Quantum Optimization, Periodicity and Tunneling"*

12:45 – 14:00 Lunch (*Maiorescu* campus restaurant)

Session 3

15:00 – 16:00 ***Gabriela Ochoa*** *(University of Stirling): "Visualising Evolution: A Network-Based Journey through Search Trajectories"*

16:00 – 16:15 Coffee break

16:15 – 18:00 PhD students: *3-minute thesis presentations*

Session 4

June 19, *Ion Simionescu* Hall

9:30 – 10:30 **Daniela Zaharie** (West University of Timişoara): *"Detecting structural changes in time series: from statistical models to Deep Learning"*

10:30 – 10:45 Coffee Break

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ECODAM 2026

FACULTY OF COMPUTER SCIENCE

BioInformatics /Applied Machine Learning track

Session 5

June 17, *Ferdinand* Hall

16:30 – 17:30 **Jung-Hsin Lin** (Academia Sinica) – *"Generative AI and Robotics for Biomedical Applications and Drug Discovery"*

Session 6

June 19, *Ion Simionescu* Hall

Satellite Workshop: Applied Machine Learning/BioInformatics

- Core Bioinformatics Group -

10:45 – 11:15 **Andi Munteanu** (PhD student University of Iasi ; RA, Core Bioinformatics group) - "*PIGEN: Genetic Algorithms in the context of Gene Regulatory Networks*"

11:15 – 11:45 **Cristian Bulgaru** (MSc student, University of Iasi; RA, Core Bioinformatics group): "*CANARD: the epigenetics angle*"

11:45 – 12:15 **Şerban Doncean** (PhD student, University of Iasi; intern, Core Bioinformatics group): "*FALCON: Gene Regulatory Network driven batch correction*"

12: 45– 14:00 Lunch (*Maiorescu* campus restaurant)

Session 7

June 19, *Ion Simionescu* Hall

15:00 – 15:30 **Ioana Hanus** (MSc student, University of Iasi; RA, Core Bioinformatics group) – *"RAVEN: graph properties overlaid on Gene Regulatory Networks*

15:30 – 16:00 **Samira Enache** (MSc student, University of Iasi; RA, Core Bioinformatics group) – *"HERON: defining covariation"*

16:00 – 16:15 Coffee break

16:15 – 17:15 **Irina Mohorianu** (Head of Bioinformatics/ Scientific Computing @ CSCI, Head of the CORE BioInformatics group, University of Cambridge): "*RoSignOL: other tools and future directions*".

Session 8

17:15 Closing session

ECODAM 2026

Abstracts

Dan Simovici (University of Massachusetts at Boston) – *"Compressibility, Mineability and Large Language Models" (*with C. Mayer and C. Saadat)

We examine data compressibility as a proxy for data minability in the classical data mining, and in the context of large language models. We focus on the influence of the parameters of LLM queries (temperature, top-k and others) on compressibility of LLM-text generation, ontology-free detection of frequent item sets by LLMs, and graph generation.

Darrell Whitley (Colorado State University) – *"Evolutionary Optimization, Quantum Optimization, Periodicity and Tunneling"*

Both evolutionary algorithms and quantum optimization methods are capable of escaping local optima through forms of "tunneling," albeit using very different mechanisms. This talk explores what quantum optimization might learn from evolutionary methods. At present, most quantum optimization algorithms rely on quantum simulated annealing. However, a quantum advantage is often achieved by exploiting quantum Fourier techniques and problem periodicity. Focusing on periodicity is likely to lead to new ways of developing quantum optimization algorithms. Evolutionary tunneling can be used to expose periodicity in the distribution of local optima.

Gabriela Ochoa (University of Stirling) – "*Visualising Evolution: A Network-Based Journey through Search Trajectories*"

This lecture provides a visually-driven introduction to evolutionary computation. We begin by illustrating the working principles of evolutionary algorithms and the concept of a search space through intuitive examples. Traditional analysis often ignores the search space and concentrates solely on the objective space; in contrast, we introduce Search Trajectory Networks (STNs) as an accessible alternative. This visual and quantitative framework maps the actual paths taken by heuristic optimisers, transforming algorithmic behaviour into graph-based structures. Through a series of case studies, we show that network analysis can reveal the hidden patterns, bottlenecks, and efficiencies of evolutionary search.

Daniela Zaharie (West University of Timişoara) – *"Detecting structural changes in time series: from statistical models to Deep Learning"*

Structural change detection in time series is a challenging problem with applications ranging from geophysical monitoring based on Remote Sensing data to EEG-based seizure prediction. This talk will provide an overview of different change point detection techniques covering frequentist and Bayesian statistics, optimization-based approaches (including evolutionary computation), and deep learning approaches. The behavior of several techniques will be illustrated for displacement time series derived from Persistent Scatterer Interferometric Synthetic Aperture Radar (PS-InSAR) data, in the context of landslide or ground subsidence event prediction.

Jung-Hsin Lin (Academia Sinica) – *"Generative AI and Robotics for Biomedical Applications and Drug Discovery"*

Artificial intelligence (AI) and novel computational methods have transformed profoundly the landscape of the drug discovery and development in almost all stages.

I will start my presentation with the structure-based design story of a small molecule drug from our collaborative team for treating neurodegenerative diseases, including Alzheimer's disease, currently in the first in human (FIH) Phase I clinical trial. I will then mention our previous methodological developments for structure-based drug discovery, for improving its computational efficiency and accuracy, and for extending its applicable domains and application scenarios. To meld to gaps between computational outcomes and experimental validation, we also created automatic laboratory systems with well-controlled experimental environment and AI robotics for carrying out cell cultures, virus titration assays, chemical synthesis, and drug testing.

Irina Mohorianu (Head of Bioinformatics/Scientific Computing @ CSCI, University of Cambridge) – "RoSignOL: *further tools and future directions*"

In the RoSignOL (Robust Signature of Life) talk I will overview three new frameworks from other members of the group – the Magpie, the SMEW and the Crecerelle one. The former are focused on spatial metabolomics and spatial transcriptomics, linked semantically via regulatory networks, the latter presents a novel perspective on transcript usage vs variation in expression, revealing an interesting, discriminative angle, able to refine subpopulations of cells. I will also discuss future directions that we will focus on over the coming months/ years, that might be relevant for prospective MSc/ PhD students.

Andi Munteanu (PhD student University of Iasi ; Intern, Core Bioinformatics group) - "*PIGEN: Genetic Algorithms in the context of Gene Regulatory Networks*"

The rapid development of CRISPR-centred single-cell experiments prompted novel perspectives on the inference and interpretation of regulatory interactions. On an AML case study from the Huntly lab (University of Cambridge), we introduce PIGEN (Perturb-seq Inference of GEne Networks), a novel framework for identifying genes impacted by single-gene perturbations, compared to non-targeted controls. The selection of genes impacted by the perturbation is optimised on a Genetic Algorithm, which is more efficient than standard identification of markers, based on log fold-changes. The covariation-based network created on impacted genes is further refined on small-world principles. Two shiny apps support the visualisation of patterns and strengthen the collaboration with wet-lab scientists.

Cristian Bulgaru (intern, Core Bioinformatics group): "*CANARD: the epigenetics angle*"

The prediction of peaks on scATACseq data is heavily influenced by the number of cells captured and overall sequencing depth; due to inherent technical variability between samples, false positives and false negatives can significantly alter analyses and bias interpretations; state-of-the-art sample-based subsampling does not fully address the issue, further biasing the quantification of chromatin openness at single-cell level. To enhance the comparability across samples, we developed CANARD (Pipeline for Assessing Co-expression of ATAC and RNA Dynamics) which is a two-step subsampling approach, iteratively optimising the number of cells and sequencing depths across cells. Following this pre-processing the peak identification, with state-of-the-art macs3, is more robust and interpretable.

Şerban Doncean (PhD student, University of Iasi; intern, Core Bioinformatics group): "*FALCON: Gene Regulatory Network driven batch correction*"

The integration of datasets is a fundamental task that relies on the comparability of transcriptomic signatures. Current approaches build on the strict mathematical alignment of distributions foregoing the interplay, and variable proportions, of signal and noise. With FALCON (Functional alignment of co-variation networks) we propose a novel perspective focused on conserved regulatory interactions and a small-world prioritisation of edges. Built in several iterations, the framework aligns the signal in a constructive manner, while also preserving the specific differences between samples. The framework will be illustrated on WT/WT and a WT/KO case study, on bulk transcriptomics.

Ioana Hanus (MSc student, University of Iasi; intern, Core Bioinformatics group) – *"RAVEN: graph properties overlaid on Gene Regulatory Networks"*

Traditional approaches for identifying differentially expressed genes rely on fold changes, abundances and inferred p-values, foregoing any information related to the functionality of genes i.e. the observed covariation patterns. With RAVEN (Robust Assessment of variation embedded in networks) we propose a regulatory prioritisation of genes on graph-based properties on inferred gene regulatory networks. On several bulk transcriptomics case studies, we underline false negatives – transcription factors, which have medium to low abundance, lower fold changes compared to non-regulatory genes but influence the variation in expression of a large number of other genes. False positives, e.g. rRNA genes are differentially expressed, but with less impact on other genes or on the topology of GRNs.

Samira Enache (MSc student, University of Iasi; intern, Core Bioinformatics group) – *"HERON: defining covariation"*

Recent studies prompted a heavier use of functional interpretations using gene regulatory networks. However, a mono-block assessment of dynamics and causality can be biased by different noise patterns, and variable regulatory strengths. To separate networks into relevant subsets, we propose the concept of GRN slicing, that identifies kernels of genes and regulatory interactions, facilitating also the characterisation of external cross-pathway interactions. The HERON framework will be illustrated on single-cell transcriptomics datasets comparing in-vivo vs in-vitro samples, as well as assessing cross patient heterogeneity.